

GRADUATION THESIS

Transfer Learning for Credit Scoring: An Experimental Study of Heterogeneous Feature Augmentation and Domain Adversarial Learning

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1.1 Research Background

- ▶ Credit scoring: turning a borrower's default likelihood into a numerical score
 - Durand (1941): first systematic statistical approach to consumer credit
 - Altman (1968): Z-score via linear discriminant analysis
 - Hand & Henley (1997): "one of the most successful applications of statistics and ML in finance"
- ▶ Post-2008: Basel III (BCBS, 2017) - banks must estimate **PD, LGD, EAD** under the Internal Ratings-Based approach
- ▶ Vietnam: CIC (1999) → model v1.0 (2015) → v2.0 (2021); most domestic banks still rely on internally developed models; 2024 SBV sandbox decree for AI/big-data credit scoring

1.2 Research Problem (1): Cold-Start and HDA

The Cold-Start Problem

Many credit institutions lack sufficient historical data on new borrowers. Large public datasets differ from the target in feature structure and distribution. Training on Dataset A and applying to Dataset B causes **negative transfer** (Pan & Yang, 2010).

Heterogeneous Domain Adaptation (HDA)

Source and target differ in both feature dimensionality and distribution: $\mathcal{X}_S \neq \mathcal{X}_T$, $d_S \neq d_T$, $P(X_S) \neq P(X_T)$. Standard DA (MMD, CORAL) requires $d_S = d_T$ - inapplicable here. HDA resolves simultaneously: (i) structural heterogeneity and (ii) distributional shift.

This thesis: LC $\rightarrow$ TDCCC

US P2P loans ($d_S \in \{120, 37\}$) as source; Taiwan credit cards ($d_T = 23$) as target. Both share $\mathcal{Y} = \{0, 1\}$ but differ in feature space, distribution, and product type.

1.2 Research Problem (2): What, Where and When to Transfer

What to Transfer - Feature Representations

Four types of knowledge: instances, feature representations, parameters, relational. Instance-based requires $d_S = d_T$; parameter-based requires shared architecture; relational requires entity graph - all three ruled out for tabular HDA. Natural choice: **feature representations**.

Where to Transfer

Source $\mathcal{D}_S$: Lending Club ($\mathbb{R}^{d_S}$, US, 2007-2018). Target $\mathcal{D}_T$: TDCCC ($\mathbb{R}^{23}$, Taiwan, 2005). Since $d_S \neq d_T$ and $P(X_S) \neq P(X_T)$: canonical HDA problem.

When to Transfer

Three conditions (Rosenstein et al., 2005) all satisfied: (i) related underlying structure (income, debt, repayment); (ii) limited target data (30K vs. 2.26M); (iii) method suits heterogeneity type ($d_S \neq d_T$).

1.2 Research Problem (3): How to Transfer and Negative Transfer

How to Transfer - Two Methods Compared

HFA (Heterogeneous Feature Augmentation) - feature mapping family: Learns two linear projections P, Q mapping source and target into a shared subspace $\mathbb{R}^{d_c}$, then augments features and trains a joint SVM classifier. Shallow, convex-per-block, CPU-friendly.

DANN (Domain-Adversarial Neural Network) - adversarial alignment family: Learns domain-invariant representations via a Gradient Reversal Layer (GRL) that compels the encoder to fool a domain classifier. Deep, non-convex, requires GPU for large runs.

Negative Transfer Risk

Negative transfer occurs when: $\varepsilon_T(\text{transfer}) > \varepsilon_T(\text{baseline})$. Three risk factors for this thesis: label shift ($P_S(Y) \neq P_T(Y)$: 14% vs. 22%), source dominance (ratio 10:1), and conceptual mismatch (loans vs. credit cards).

Prevention: 3-case ablation protocol.

1.3 Research Questions

RQ 1: To what extent can transfer learning (HFA or DANN) improve credit scoring performance on TDCCC when Lending Club is used as the source domain, compared to a baseline trained only on the target?

RQ 2: Which of the two approaches - HFA or DANN - shows greater performance for credit scoring, and under what conditions do their strengths or weaknesses become most apparent?

RQ 3: How do specific design choices shape the effectiveness of transfer learning, as revealed through the ablation experiments?

RQ 4: What risks of negative transfer emerge when applying HDA across domains with different markets, products, and label distributions, and how can experimental protocols detect and mitigate these risks?

2.1 Transfer Learning: Taxonomy and Key Concepts

Taxonomy (Pan & Yang, 2010)

Transfer learning classified along three dimensions: (i) source and target domains identical?, (ii) source and target tasks identical?, (iii) target domain labeled? Four main classes: inductive, transductive, unsupervised, negative. Weiss et al. (2016) extended to deep learning and DA methods.

Heterogeneous vs. Homogeneous DA

Homogeneous DA: $d_S = d_T$, only $P(X)$ shifts - standard MMD, CORAL applicable.

Heterogeneous DA: $d_S \neq d_T$ - requires specialized mapping methods.

HFA and DANN (with domain-specific encoders) were developed precisely for this case.

Key reference: Yang, Zhang, Dai & Pan (2020), *Transfer Learning*, Cambridge University Press - the primary reference for this thesis (HFA in Chapter 6, DANN in Chapter 8).

2.1 H-divergence Theory (Ben-David et al., 2010)

Target risk bound (Theorem 2):

$$\varepsilon_T(h) \leq \varepsilon_S(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) + \lambda^*$$

$\varepsilon_T(h)$: target risk. $\varepsilon_S(h)$: source risk. $d_{\mathcal{H}\Delta\mathcal{H}}$: H-divergence between domains. $\lambda^* = \min_h [\varepsilon_S(h) + \varepsilon_T(h)]$: ideal joint hypothesis error.

Two key implications

- (1) Reducing $d_{\mathcal{H}\Delta\mathcal{H}}$ tightens the bound on target risk - the motivation behind DANN's adversarial training.
- (2) If λ^* is large (dissimilar domains), the bound becomes vacuous - theoretical root cause of negative transfer in this thesis.

Warning (Zhao et al., 2019): minimising adversarial loss does not guarantee minimising H-divergence when $P_S(Y) \neq P_T(Y)$ - directly applicable here since LC default rate ($\approx 14\%$) $\neq$ TDCCC ($\approx 22\%$).

2.2 Overview of Credit Scoring

Classical Statistical Era (1940-1990): Durand (1941): first systematic statistical approach. Altman (1968): Z-score model (5 ratios, 95% bankruptcy accuracy). Cox (1958), Eisenbeis (1977): logistic regression becomes industry standard.

Classical ML Era (1990-2010): Baesens et al. (2003): SVM with RBF kernel as strong baseline (8 datasets). Yeh & Lien (2009): introduced TDCCC, compared 6 methods. Hand (2009): "measure problem" - metric choice changes model ranking.

Modern ML and Deep Learning (2010-present): Lessmann et al. (2015): 41 classifiers - RF and GBM improve AUC by 1-3 pp over logit. Borisov et al. (2024): XGBoost/LightGBM match or outperform deep models on tabular data. Deep gains only with self-supervised pretraining or transfer learning - precisely the setting explored in this thesis.

2.3 Transfer Learning for Credit Scoring and Research Gap

Homogeneous: Tsai & Wu (2008): multi-source with shared schema. Bracke et al. (2019): GBM on LC with explainability.

Heterogeneous: Few studies assume $d_S \neq d_T$. Xu & Zhang (2025): adversarial transfer for internet finance. This thesis: one of the few HeteroDANN applications to tabular credit data.

Negative Transfer: Rosenstein et al. (2005): transfer not always beneficial. Pan & Yang (2010, §3): avoidance via source selection, reweighting. Wang et al. (2019): 3 root causes - large domain gap, poor model class, size imbalance.

Three Research Gaps Addressed

- (1) HDA for credit scoring understudied - most assume $d_S = d_T$.
- (2) Shallow vs. deep HDA rarely compared on tabular financial data.
- (3) No rigorous 3-case ablation benchmark with reproducibility in credit scoring.

3.1 Target Domain: Taiwan Default of Credit Card Clients

Taiwan Default of Credit Card Clients (Yeh & Lien, 2009)

30,000 records, April-September 2005, UCI Repository. Default rate: **22.12%**

Group	Key Features	Description
Continuous (14)	LIMIT_BAL, BILL_AMT1-6, PAY_AMT1-6	Credit limit, monthly bill/payment amounts
Nominal (3)	SEX, EDUCATION, MARRIAGE	Demographics
Ordinal (6)	PAY_0, PAY_2-6	Repayment status (-1 to 9)
Label	default.payment.next.month	Binary: 1 = default

Preprocessing: EDUCATION/MARRIAGE: undocumented codes remapped to "Others" per codebook. PAY_X: clipped to $[-2, 8]$ (Yeh, 2009, §4.1). Type casting applied. Result: $\mathbf{d_T = 23}$ features.

3.2 Source Domain: Lending Club - Preprocessing

Lending Club (US P2P, 2007-2018)

2,260,668 records, 145 features. Final subsample: 300,000 records.

Default rate: **21.39%**

6-step preprocessing pipeline (key steps):

1. Remove 38 high-missingness cols ($> 90\%$) and 23 post-origination leakage cols
2. Label: Charged Off/Late $\rightarrow y = 1$; Fully Paid $\rightarrow y = 0$; exclude In-Progress
3. Imputation: median/mode ($< 10\%$ missing); "unknown" sentinel (10-90%)
4. Encoding: binary (5 cols), label-encode 3 ordinals, one-hot 4 nominals
5. Cleanup: drop residual object columns $\rightarrow d_S = 120$ features
6. Stratified subsampling $\rightarrow$ 300,000 records, 21.39% default rate

3.2 Lending Club: Feature Selection (2 Cases)

Case 1 (Full): $d_S = 120$ features.

Case 2 (Selected): $d_S = 37$ features guided by 5 C's of Credit (Altman & Saunders, 1997) for semantic alignment with TDCCC.

Criterion	LC features	TDCCC features
Character	delinq_2yrs, grade, sub_grade, cr_history_yr	PAY_0 - PAY_6
Capacity	annual_inc, dti, emp_length, installment	LIMIT_BAL, PAY_AMT1-6
Capital	loan_amnt, revol_util, open_acc, total_acc	BILL_AMT1-6, LIMIT_BAL
Collateral	home_ownership_* (one-hot)	MARRIAGE
Conditions	int_rate, term, purpose_*, verification_status_*	EDUCATION

Rationale: 69% feature reduction (120 $\rightarrow$ 37)

Aligned features mitigate negative transfer under large domain gap.

3.3 Domain Comparison and Heterogeneity Analysis

Criterion	TDCCC (Target)	Lending Club (Source)
Country	Taiwan	United States
Period	Apr-Sep 2005	2007-2018
Product	Credit card (revolving)	Term loan (36/60 months)
Sample size	30,000	$\approx 2.26\text{M}$ (sub. to 300K)
Feature dim.	$d_T = 23$	$d_S = 120$ (C1) / 37 (C2)
Default rate	$\approx 22\%$	$\approx 14\%$
Currency	TWD	USD
Default definition	Missed next-month payment	Charged off / Default

3.3 Domain Comparison and Heterogeneity Analysis

Four dimensions of heterogeneity:

- ▶ **Feature space:** $\mathcal{X}_S \neq \mathcal{X}_T$, $d_S \neq d_T$ - HDA by definition
- ▶ **Label shift:** $P_S(Y) \neq P_T(Y)$ (14% vs. 22%) - H-divergence bound may be non-tight
- ▶ **Covariate shift:** currency, income scale (Taiwan \$15K vs. US \$65K median), credit culture
- ▶ **Concept shift:** $P_S(Y|X) \neq P_T(Y|X)$ - TDCCC predates LC by ≥ 2 years (temporal inversion), different default definitions

4.1 Baseline Models

Three baselines representing three paradigms (Lessmann et al., 2015):

Model	Key parameters
Logistic Regression	$C = 0.1$ (stronger L2), LBFGS solver, max_iter=1000, class_weight="balanced", random_state=42
Random Forest	n_estimators=200, max_depth=8, class_weight="balanced", n_jobs=-1, random_state=42
XGBoost	n_estimators=200, max_depth=4, learning_rate=0.05, scale_pos_weight=3.521 (= 18691/5309), eval_metric="logloss", random_state=42

Data split: 80/20 stratified (seed=42)

- ▶ Train pool: 24,000 samples (default rate = 22.12%)
- ▶ Hold-out test: 6,000 samples (default rate = 22.12%)

4.1 Evaluation Metrics

AUC-ROC (Fawcett, 2006): threshold-independent discrimination. Equivalent to the probability that the model ranks a random positive above a random negative:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u)) \, du = P(\hat{p}(x^+) > \hat{p}(x^-))$$

$\text{AUC} \geq 0.70$: acceptable under Basel III (BCBS, 2017, §4.5).

KS Statistic: maximum distance between good/bad CDFs, standard banking metric: $\text{KS} = \sup_t |F_{\text{good}}(t) - F_{\text{bad}}(t)|$

4.1 Evaluation Metrics

Brier Score (Brier, 1950): calibration quality:

$BS = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2 \in [0, 1]$, lower is better.

KS-optimal threshold (Youden, 1950): used for Sensitivity, Specificity, Accuracy:

$$\tau^* = \arg \max_{\tau} J(\tau), \quad J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1$$

Krzanowski & Hand (2009): $\max_{\tau} J(\tau) = \text{KS}$. Threshold determined on training fold/pool, applied to test - no leakage.

4.2 HFA: Theoretical Foundation

Core idea (Duan et al., 2012; Yang et al., 2020, Ch. 6): Since $d_S \neq d_T$, HFA maps each sample into an augmented vector of three blocks:

$$\varphi_s(x_s) = \begin{bmatrix} Px_s \\ x_s \\ \mathbf{0}_{d_T} \end{bmatrix} \in \mathbb{R}^{d_c+d_S+d_T}, \quad \varphi_t(x_t) = \begin{bmatrix} Qx_t \\ \mathbf{0}_{d_S} \\ x_t \end{bmatrix} \in \mathbb{R}^{d_c+d_S+d_T}$$

$P \in \mathbb{R}^{d_c \times d_S}$, $Q \in \mathbb{R}^{d_c \times d_T}$: learned linear projections. Block 1 (Px_s / Qx_t): shared subspace for cross-domain comparison. Block 2/3 (original features + zero-padding): preserve domain-specific information. After augmentation, all points reside in $\mathbb{R}^{d_c+d_S+d_T}$, enabling a joint SVM.

Key inner product property:

$$\varphi_s^\top(x_s) \cdot \varphi_t(x_t) = x_s^\top P^\top Q x_t$$

All source-target interaction is mediated through $P^\top Q$. Zero-padding ensures raw blocks do not interact directly - knowledge transfer is controlled entirely through the shared subspace.

4.2 HFA: Objective Function (Yang et al., 2020, Eq. 6.8)

HFA optimization problem:

$$\min_{P, Q, \mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_P}{2} \|P\|_F^2 + \frac{\lambda_Q}{2} \|Q\|_F^2 + C \sum_{i=1}^{n_s+n_t} \xi_i^2$$

$$\text{s.t. } y_i(\mathbf{w}^\top \varphi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0, \|P\|_F \leq r_P, \|Q\|_F \leq r_Q$$

Role of each term:

- ▶ $\frac{1}{2} \|\mathbf{w}\|^2$: L2 regularization for the SVM hyperplane
- ▶ $\frac{\lambda_P}{2} \|P\|_F^2, \frac{\lambda_Q}{2} \|Q\|_F^2$: soft Frobenius regularization
- ▶ $C \sum \xi_i^2$: total squared hinge loss (Chapelle, 2007)
- ▶ $\|P\|_F \leq r_P, \|Q\|_F \leq r_Q$: Frobenius ball - guards against negative transfer by limiting the shared block's dominance over the raw feature blocks

Frobenius projection (Boyd & Vandenberghe, 2004):

$$\Pi_r(M) = M \cdot \min\left(1, \frac{r}{\|M\|_F}\right), \text{ closed-form solution, computationally cheap.}$$

4.2 HFA: Block Coordinate Descent Algorithm

Why BCD? (Tseng, 2001, Theorem 4.1)

Eq. 6.8 is not jointly convex in $(P, Q, \mathbf{w})$ but is convex in each block when others are fixed.

BCD guarantees every accumulation point is a stationary point under three conditions:

- (i) Continuity and per-block convexity,
- (ii) Cartesian product feasible set (Frobenius ball $\times \mathbb{R}^d$),
- (iii) Non-increasing objective at each sub-step.

4.2 HFA: Block Coordinate Descent Algorithm

Algorithm (as implemented in `train_hfa`):

- 1. Initialization:** $P_0, Q_0 \sim \mathcal{N}(0, 1/d_c)$ (Johnson-Lindenstrauss), immediately projected onto Frobenius ball
- 2. For** $k = 1, \dots, n_{\text{outer}} (= 5)$:
 - 1** Fix (P_{k-1}, Q_{k-1}) : augment features, fit LinearSVC $\rightarrow$ update $(\mathbf{w}_k, b_k)$
 - 2** Fix $(\mathbf{w}_k, b_k)$: compute subgradients g_P, g_Q , apply projected gradient step: $P_k \leftarrow \Pi_{r_P}(P_{k-1} - \eta g_P)$, $Q_k \leftarrow \Pi_{r_Q}(Q_{k-1} - \eta g_Q)$

JL initialization:

$\Pr[(1 - \varepsilon)\|x\|^2 \leq \|P x\|^2 \leq (1 + \varepsilon)\|x\|^2] \geq 1 - 2 \exp(-d_c \varepsilon^2 / 8)$. For $d_c = 64$, $\varepsilon = 0.5$: probability ≥ 0.73 .

4.2 HFA: Subgradient and Platt Calibration

Squared-hinge: $\ell(z) = \max(0, 1 - z)^2$, $\ell'(z) = -2 \max(0, 1 - z)$.

Differentiable everywhere, enabling smooth BCD convergence.

Gradient w.r.t. P :

$$g_P = -2C \sum_{i \in \mathcal{A}_S} \max(0, 1 - y_i z_i) \cdot y_i \cdot \mathbf{w}_c x_{S,i}^\top + \lambda_P P$$

$\mathbf{w}_c$: common-block portion of $\mathbf{w}$. $\mathcal{A}_S = \{i : 1 - y_i z_i > 0\}$: active set (margin violators). Analogously for g_Q over $\mathcal{A}_T$.

Projected gradient update: $P_k \leftarrow \Pi_{r_P}(P_{k-1} - \eta g_P)$,
 $Q_k \leftarrow \Pi_{r_Q}(Q_{k-1} - \eta g_Q)$.

Platt calibration (Platt, 1999):

$$\hat{P}(y = 1|x) = \frac{1}{1 + \exp(-(Af(x) + B))}$$

(A, B) estimated by MLE: `LogisticRegression( $C=10^6$ , balanced)` on calibration set.

4.2 HFA: Experimental Parameters

Hyperparameter	Symbol	Value	Rationale
Common space dim.	d_c	64	JL preservation prob. ≥ 0.73 at $\varepsilon = 0.5$
BCD outer iterations	n_{outer}	5	Sufficient (Duan et al., 2012: 3-10 typical)
Subgradient LR	η	10^{-3}	Avoids overshooting Frobenius boundary
SVM regularization	C	1.0	Standard LinearSVC default
Soft Frobenius reg.	λ_P, λ_Q	10^{-3}	Light; hard constraint is primary control
Hard Frobenius radius	r_P, r_Q	10.0	Bounds projection magnitude
SVM class weight	-	balanced	Corrects class imbalance ($\approx 22\%$)
SVM max_iter	-	2000	Ensures internal convergence
Platt calibration C	-	10^6	Near-zero regularization $\approx$ pure MLE
Source sample size	-	300,000	Stratified subsample from 1,325,535

4.3 DANN: H-divergence and Min-Max Objective

DANN operationalizes the H-divergence bound through adversarial training. The two levers for reducing target risk: (i) reduce $\varepsilon_S(h)$ via supervised learning on source; (ii) reduce $d_{\mathcal{H}\Delta\mathcal{H}}$ by making representations indistinguishable between domains.

DANN min-max objective (Ganin et al., 2016):

$$\min_{G_f, G_y} \max_{G_d} \mathcal{L}_y(G_y(G_f(x_s)), y_s) - \lambda \cdot \mathcal{L}_d(G_d(G_f(x)), d)$$

- ▶ G_f : feature extractor (encoder) - maps inputs to $\mathbb{R}^{128}$
- ▶ G_y : label predictor - predicts default probability from representation
- ▶ G_d : domain classifier - distinguishes source ($d=0$) from target ($d=1$)
- ▶ λ : adversarial balancing coefficient (warm-up schedule)

At Nash equilibrium: encoder produces representations that domain classifier cannot distinguish $\rightarrow$ domain accuracy $\rightarrow 0.5$ (random guess) $\rightarrow d_{\mathcal{H}\Delta\mathcal{H}} \rightarrow 0$. This is an **unsupervised** DA method: only source labels y_s used during training; target labels y_t reserved for evaluation.

4.3 DANN: GRL and HeteroDANN Architecture

Gradient Reversal Layer (GRL) - solves min-max in a single backward pass:

$$\text{Forward: } x \mapsto x, \quad \text{Backward: } \frac{\partial \mathcal{L}}{\partial x} \mapsto -\lambda \cdot \frac{\partial \mathcal{L}}{\partial x}$$

G_d : receives unmodified gradients - learns to distinguish domains. G_f : receives sign-reversed gradients through GRL - learns to fool G_d . G_y : bypasses GRL, optimizes classification loss.

HeteroDANN - two domain-specific encoders (separate input dimensions):

Module	Architecture	Output
Encoder_S ($d_S \in \{120, 37\}$)	Linear $\rightarrow$ LN $\rightarrow$ ReLU $\rightarrow$ Drop(0.2) $\rightarrow$ Linear $\rightarrow$ LN $\rightarrow$ ReLU	$\mathbb{R}^{128}$
Encoder_T ($d_T = 23$)	Linear $\rightarrow$ LN $\rightarrow$ ReLU $\rightarrow$ Drop(0.2) $\rightarrow$ Linear $\rightarrow$ LN $\rightarrow$ ReLU	$\mathbb{R}^{128}$
LabelPredictor	Linear(128,32) $\rightarrow$ ReLU $\rightarrow$ Linear(32,1)	logit
DomainClassifier	GRL $\rightarrow$ Linear(128,32) $\rightarrow$ ReLU $\rightarrow$ Linear(32,1)	logit

LayerNorm (not BatchNorm): batch contains mixed-domain samples; LN normalizes per-sample, independent of batch composition.

4.3 DANN: Training Schedules

Lambda warm-up (Ganin et al., 2016, §4.2):

$$\lambda(p) = \frac{2}{1 + \exp(-10p)} - 1, \quad p = \frac{\text{step}}{\text{total_steps}}$$

$\lambda(0) = 0$: no adversarial signal at start. $\lambda(0.5) \approx 0.987$: rapid increase.
 $\lambda(1) \approx 1$: full adversarial pressure at end.

LR polynomial decay: $\eta(p) = \eta_0 / (1 + 10p)^{0.75}$, $\eta_0 = 0.01$.

Combined loss at each step:

$$\mathcal{L} = \mathcal{L}_s(\hat{y}_s, y_s) + \lambda \cdot \mathcal{L}_d(\hat{d}, d)$$

$\mathcal{L}_s$: BCEWithLogitsLoss on source; $\text{pos_weight} = N_{\text{neg}}/N_{\text{pos}} \approx 6.14$. $\mathcal{L}_d$: BCEWithLogitsLoss on all samples (source=0, target=1).

Training parameters: total_steps=4000, batch_size=64, hidden_dim=128, Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$), gradient clipping $\|\nabla\|_2 \leq 1.0$ (Pascanu et al., 2013).

4.4 Negative Transfer Ablation - Design

3-Case Protocol

Isolates each component's contribution; tests for negative transfer explicitly. Applied to both HFA and DANN on the 6,000-sample hold-out test set.

Case	Description
Case 1 - Target only	Trained exclusively on TDCCC; no LC data. HFA uses LinearSVC; DANN uses Encoder_T + LabelPredictor only. Method-specific baseline.
Case 2 - Naive concat	LC + TDCCC combined without alignment. Block-diagonal concat for HFA; zero-padding for DANN. Tests source data alone.
Case 3 - Full method	Complete transfer: HFA with BCD + Frobenius, or DANN with GRL + adversarial training. Tests alignment mechanism benefit.

4.4 Negative Transfer Ablation - Meaning

Interpretation:

- ▶ C3 vs. C1: net transfer benefit.
- ▶ C3 vs. C2: alignment mechanism contribution.
- ▶ C2 vs. C1: raw source data effect.
- ▶ $C3 < C1$: negative transfer confirmed.

5.1 Baseline Results: 5-Fold Cross-Validation

5-fold stratified CV on train pool (24,000 samples):

Model	AUC	KS	Brier	Sens.	Spec.	Acc.
Logistic Regression	0.7262	0.3737	0.2058	0.5413	0.8323	0.7680
Random Forest	0.7790	0.4281	0.1756	0.6387	0.7894	0.7561
XGBoost	0.7819	0.4319	0.1771	0.6257	0.8062	0.7662

Observations:

- ▶ All three models achieve stable and consistent performance across folds
- ▶ XGBoost leads on AUC (0.7819) - gradient boosting optimizes bias reduction
- ▶ Random Forest leads on KS (0.4281) - bagging optimizes variance reduction
- ▶ Logistic Regression achieves highest Specificity (0.8323) - most conservative
- ▶ XGBoost selected as the reference baseline (highest AUC)

5.1 Baseline Results: Hold-out Evaluation and Analysis

Hold-out evaluation (6,000 samples):

Model	AUC	KS	Brier	Sens.	Spec.	Acc.
Logistic Regression	0.7084	0.3586	0.2089	0.5207	0.8376	0.7675
Random Forest	0.7739	0.4333	0.1786	0.6420	0.7734	0.7443
XGBoost	0.7758	0.4235	0.1801	0.6586	0.7541	0.7330

Confusion matrix (hold-out):

Model	TP	TN	FP	FN
Logistic Regression	691	3914	759	636
Random Forest	852	3614	1059	475
XGBoost	874	3524	1149	453

CV-holdout gap: XGBoost $\Delta\text{AUC} = 0.0061$; RF $\Delta\text{AUC} = 0.0051$ - no substantial overfitting. Results consistent with Yeh & Lien (2009): AUC ≈ 0.77 on TDCCC. **XGBoost (AUC = 0.7758)**: any transfer method below this threshold is treated as evidence of negative transfer.

5.2 HFA Results: 5-Fold Cross-Validation

Case 1 (Full LC, $d_S = 120$):

Fold	AUC	KS	Brier	Sens.	Spec.	Acc.	Thresh.
1	0.7287	0.3906	0.2061	0.5287	0.8561	0.7837	0.568
2	0.7205	0.3895	0.2042	0.5659	0.8058	0.7527	0.530
3	0.7078	0.3589	0.2101	0.5395	0.8133	0.7527	0.538
4	0.7088	0.3644	0.2090	0.5009	0.8585	0.7794	0.576
5	0.7213	0.3754	0.2058	0.4953	0.8777	0.7931	0.581
Mean	0.7174 ± 0.0089	0.3758 ± 0.014	0.2070 ± 0.002	0.5261 ± 0.029	0.8423 ± 0.031	0.7723 ± 0.019	-

Case 2 (Selected LC, $d_S = 37$) - Mean results:

AUC	KS	Brier	Sensitivity	Specificity	Accuracy
0.7218 ± 0.011	0.3814 ± 0.015	0.2062 ± 0.002	0.5438 ± 0.025	0.8272 ± 0.027	0.7646 ± 0.016

Very low variance (± 0.009 - 0.011) confirms **stable and reliable** behavior across folds. Case 2 AUC: $+0.44$ pp over Case 1, with consistent low std across all metrics.

5.2 HFA Results: Hold-out and Analysis

Hold-out results vs. baseline:

Method	AUC	KS	Brier	Sens.	Spec.
HFA Case 1 (Full LC)	0.6993	0.3613	0.2100	0.5011	0.8571
HFA Case 2 (Selected)	0.7042	0.3595	0.2092	0.5290	0.8226
XGBoost (reference)	0.7758	0.4235	0.1801	0.6586	0.7541
LinearSVC on TDCCC	0.7076	0.3572	0.2089	0.5162	0.8395

Key findings

HFA below XGBoost: -7.65 pp (C1), -7.16 pp (C2) - **negative transfer confirmed**. Also below LinearSVC on TDCCC alone (-0.83 pp C1), confirming source data did not help.

Notable: Specificity exceeds XGBoost (0.8571 vs. 0.7541); 69% feature reduction adds only $+0.49$ pp AUC; very low variance (± 0.009) - stable and deployable.

5.3 DANN Results: 5-Fold Cross-Validation

Case 1 (Full LC, $d_S = 120$) and Case 2 (Selected, $d_S = 37$): Case 2 results are identical to Case 1 (same target domain $d_T = 23$ drives the outcome).

Fold	AUC	KS	Brier	Sensitivity	Specificity
1	0.4222	0.0083	0.3363	0.9689	0.0348
2	0.4707	0.0559	0.2473	0.3785	0.6747
3	0.6009	0.2007	0.2510	0.5659	0.6281
4	0.6454	0.2769	0.4710	0.4680	0.8066
5	0.4165	0.0408	0.2783	0.9388	0.0928
Mean	0.5111 ± 0.0944	0.1165 ± 0.104	0.3168 ± 0.093	0.664 ± 0.273	0.447 ± 0.357

Critical observation

AUC ≈ 0.50 -0.51: equivalent to a random classifier. KS < 0.12 : below accepted minimum (KS ≥ 0.20) in credit scoring. Std ± 0.10 : ≈ 12 times larger than HFA (± 0.009). Fold 1: Sensitivity = 0.969, Specificity = 0.035 - complete model collapse.

5.3 DANN Results: Failure Mechanism Analysis

GRL Gradient Saturation (the core failure mechanism)

Training logs confirm: $\mathcal{L}_d \approx 0$ from step ~ 200 onwards. This means the domain classifier separates source from target with near-perfect confidence - NOT failing to learn.

Why saturation occurs: Two encoders with different input dimensions ($d_S = 120$ vs. $d_T = 23$) produce embeddings that, despite compression to $\mathbb{R}^{128}$, retain enough dimension-specific signal for the domain classifier to separate trivially.

Cascade: (1) $\sigma(\text{logit}_d) \approx 1 \rightarrow$ gradient through GRL ≈ 0 . (2) Encoder receives no adversarial signal. (3) $\mathcal{L}_s$ optimized on source alone $\rightarrow$ representations fail on target. (4) Case 3 collapses to AUC = 0.4655 (Full LC) / 0.6167 (Selected) - near random.

This is **gradient saturation under heterogeneous adversarial alignment**. DANN is optimisation-hard (not ill-posed) when $d_S \gg d_T$.

5.4 Ablation Study: HFA

HFA 3-case ablation on hold-out (6,000 samples):

Case 1 - Full LC ($d_S = 120$):

Case	AUC	KS	Brier	Sens.	Spec.	Thresh.
C1 - Target only	0.7076	0.3572	0.2089	0.5162	0.8395	0.554
C2 - Naive concat	0.7077	0.3571	0.2128	0.6405	0.6653	0.498
C3 - Full HFA	0.6993	0.3613	0.2100	0.5011	0.8571	0.570

Case 2 - Selected LC ($d_S = 37$):

Case	AUC	KS	Brier	Sens.	Spec.	Thresh.
C1 - Target only	0.7076	0.3572	0.2089	0.5162	0.8395	0.554
C2 - Naive concat	0.7077	0.3576	0.2132	0.6458	0.6574	0.496
C3 - Full HFA	0.7042	0.3595	0.2092	0.5290	0.8226	0.541

HFA achieves **safe transfer**: $\Delta\text{AUC} < 0.009$, and Full HFA improves KS and Specificity over naive concatenation - alignment has a positive effect on threshold calibration and false positive control even under large gap.

5.4 Ablation Study: DANN

DANN 3-case ablation on hold-out (6,000 samples):

Case 1 - Full LC ($d_S = 120$):

Case	AUC	KS	Brier	Sens.	Spec.	Thresh.
C1 - Target only	0.7659	0.4013	0.1841	0.6345	0.7625	0.449
C2 - Naive concat	0.6837	0.3080	0.2393	0.4431	0.8615	0.612
C3 - Full DANN	0.4655	0.0530	0.1949	0.7476	0.3019	0.124

Case 2 - Selected LC ($d_S = 37$):

Case	AUC	KS	Brier	Sens.	Spec.	Thresh.
C1 - Target only	0.7681	0.4012	0.1884	0.6451	0.7526	0.458
C2 - Naive concat	0.6814	0.3080	0.2314	0.4476	0.8575	0.600
C3 - Full DANN	0.6167	0.2095	0.3281	0.4928	0.7143	0.649

DANN Case 1 - target only (MLP only, no LC data) achieves AUC = 0.7659-0.7681, surpassing XGBoost. Full DANN (Case 3) collapses: -0.30 AUC drop on Case 1, -0.15 on Case 2.

5.4 Cross-Method Ablation Comparison

Method	Feature Set	C1 AUC	C2 AUC	C3 AUC	Trend C1→C3
HFA	Full LC	0.7076	0.7077	0.6993	↓ small (−0.0083)
HFA	Selected	0.7076	0.7077	0.7042	↓ tiny (−0.0034)
DANN	Full LC	0.7659	0.6837	0.4655	↓↓ large (−0.3004)
DANN	Selected	0.7681	0.6814	0.6167	↓ big (−0.1514)

Conclusion

HFA achieves **safe transfer** ($\Delta\text{AUC} < 0.009$). DANN shows **severe negative transfer** (ΔAUC up to -0.30).

Critical design implication - Case 1 vs. Case 2 for DANN:

DANN gain from Case 1 to Case 2: +15.12 pp (0.4655 → 0.6167). HFA gain: only +0.49 pp. Semantically guided feature reduction plays a critical role when the architecture relies on a shared representation space - an important design implication for future HDA systems.

5.5 Overall Comparison: Method Performance

Method	Set	AUC	KS	Brier	Sensitivity	Specificity
Logistic Regression	-	0.7084	0.3586	0.2089	0.5207	0.8376
Random Forest	-	0.7739	0.4333	0.1786	0.6420	0.7734
XGBoost (best ref.)	-	0.7758	0.4235	0.1801	0.6586	0.7541
HFA Full	C1	0.6993	0.3613	0.2100	0.5011	0.8571
HFA Full	C2	0.7042	0.3595	0.2092	0.5290	0.8226
DANN Full	C1	0.4655	0.0530	0.1949	0.7476	0.3019
DANN Full	C2	0.6167	0.2095	0.3281	0.4928	0.7143

- ▶ XGBoost: AUC = 0.7758 - highest overall, reference benchmark
- ▶ HFA below XGBoost: -7.65 pp (C1), -7.16 pp (C2) - mild negative transfer
- ▶ DANN below XGBoost: -31.03 pp (C1), -15.91 pp (C2) - severe negative transfer
- ▶ HFA Specificity exceeds XGBoost (0.8571 vs. 0.7541 on C1) - better false positive control
- ▶ Both methods exhibit negative transfer from the large domain gap

5.5 Overall Comparison: Feature Set Impact and Stability

Feature set impact (ΔAUC : Case 2 - Case 1):

Method	ΔAUC	Mechanism
HFA	+0.49 pp	Fewer noisy features reduce BCD projection error
DANN	+15.12 pp	Semantic alignment narrows gap, $\mathcal{L}_d$ escapes collapse

5-fold CV stability:

Method	Feature Set	AUC (CV)	Std
HFA	C1	0.7174	± 0.0080
HFA	C2	0.7218	± 0.0098
DANN	C1	0.5048	± 0.1103
DANN	C2	0.5111	± 0.0944

Stability finding

DANN std is $\approx 12\text{-}14\times$ larger than HFA (± 0.10 vs. ± 0.009): structural instability, not random noise. HFA's low variance is critical in production credit scoring (Siddiqi, 2006).

5.5 Root Cause Analysis of Negative Transfer

Three root causes (Ben-David et al., 2010):

- 1. Excessive domain gap (λ^* too large):** LC (P2P lending, US, 2007-2018) vs. TDCCC (credit cards, Taiwan, 2005). Different default dynamics, credit culture, currency. Source (2007-2018) postdates target (2005) by ≥ 2 years - temporal inversion. Dimensionality ratio $d_S/d_T \approx 5.2$ reflects structural asymmetry.
- 2. Linear constraint (HFA-specific):** Linear mappings inadequate when distributional differences are nonlinear. Pan et al. (2011): effective DA requires jointly reducing marginal and conditional distributions. Frobenius constraint $\|P\|_F \leq r_P$ may be too restrictive for the large domain gap.
- 3. GRL gradient saturation (DANN-specific):** $\sigma(\text{logit}_d) \approx 1 \rightarrow$ gradient through GRL $\approx 0 \rightarrow$ encoder receives no adversarial signal $\rightarrow \mathcal{L}_s$ on source alone. Zhao et al. (2019): forcing H-divergence to zero under label shift destroys the joint distribution.

6.1 Research Summary

Experimental pipeline

1. Preprocessing: two strategies - Case 1 ($d_S = 120$) and Case 2 ($d_S = 37$, 5 C's-guided)
2. Three baselines: LR / RF / XGBoost (reference AUC = 0.7758)
3. HFA and DANN trained and evaluated via 5-fold CV + independent hold-out
4. 3-case ablation to detect negative transfer explicitly

Main result: Neither transfer method surpasses the strongest baseline.

	HFA	DANN
Hold-out AUC (Full, C2)	0.7042	0.6167
vs. XGBoost reference	-7.16 pp	-15.91 pp
CV AUC (C2)	0.7218 ± 0.0098	0.5111 ± 0.0944
Negative transfer	Mild, safe	Severe collapse
Failure mode	Linear constraint	GRL degeneracy

6.2 HFA Evaluation: Advantages and Limitations

Advantages: Convex-per-block with convergence guarantee (Tseng, 2001, Th. 4.1); CPU-friendly. Low variance (± 0.009) - production reliability. Specificity exceeds XGBoost (0.8571 vs. 0.7541). BCD converges within 5 iterations; Frobenius projection in closed form.

Limitations: Linear mappings only - cannot capture nonlinear distributional differences. Sensitive to initial subspace quality. Frobenius constraint may be too restrictive; mild negative transfer under large domain gap.

Conclusion

HFA achieves **safe transfer** on LC-TDCCC. AUC -7 pp vs. XGBoost, but low variance and stable - appropriate when deployment reliability matters as much as absolute discriminative performance.

6.3 DANN Evaluation: Advantages and Limitations (1)

Advantages of DANN:

- ▶ Deep encoder captures complex nonlinear relationships (Case 1 hold-out AUC ≈ 0.77 , surpassing XGBoost - architecture is strong)
- ▶ GRL is elegant: only a single backward pass for the min-max optimization
- ▶ Warm-up and polynomial LR decay smoothly control adversarial signal strength
- ▶ Can learn domain-invariant representations effectively when $d_S \approx d_T$

Limitations of DANN:

- ▶ Extremely unstable when encoders have different input dimensions ($d_S \neq d_T$)
- ▶ H-divergence bound not directly applicable under label shift ($P_S(Y) \neq P_T(Y)$)
- ▶ Large fold-to-fold variance (± 0.10) - unreliable in production
- ▶ GRL gradient saturation degrades performance to near-random in the heterogeneous setting

6.3 DANN Evaluation: Advantages and Limitations (2)

Conclusion

DANN's observed benefit in Case 1 ($\text{AUC} \approx 0.77$) comes from the MLP architecture, not the adversarial alignment mechanism - which is effectively disabled by GRL degeneracy. For the LC-TDCCC heterogeneous domain pair, linear feature alignment (HFA) is the safer and more reliable choice.

Key distinction from HFA:

- ▶ HFA's negative transfer is mild (-7 pp AUC) and stable (± 0.009 std)
- ▶ DANN's negative transfer is severe (-31 pp AUC on C1) and unstable (± 0.10 std)
- ▶ DANN's CASE 1 target-only result ($\text{AUC} \approx 0.77$) shows the MLP architecture has real capacity - but this benefit disappears once the GRL is activated
- ▶ The failure is structural: GRL gradient saturation is a fundamental incompatibility between the adversarial alignment mechanism and heterogeneous feature spaces

6.4 HFA vs DANN Comparison

Criterion	HFA	DANN
Model type	Linear (P, Q + LinearSVC)	Nonlinear (MLP + GRL)
AUC hold-out (Full, C2)	0.7042	0.6167
KS hold-out (Full, C2)	0.3595	0.2095
Δ vs. Case 1 (target only)	-0.34 pp	-15.14 pp
AUC 5-fold CV (C2)	0.7218 ± 0.0098	0.5111 ± 0.0944
Convergence guarantee	Yes (Tseng, 2001, Th. 4.1)	No (non-convex)
Training cost	Low (CPU)	Moderate (CPU/GPU)
Interpretability	High (linear)	Low (deep)
Robustness under label shift	Low	Low (GRL collapse)
Fold-to-fold variance	Low (± 0.009)	High (± 0.094)

On LC $\rightarrow$ TDCCC: HFA outperforms DANN (AUC 0.7042 vs. 0.6167; variance ≈ 10 times lower), though neither surpasses XGBoost baseline.

6.5 Limitations

1. **Computational constraints:** HFA tested with $d_c \in \{24, 64\}$; DANN with `hidden_dim = 128`. Larger subspaces or deeper encoders untested.
2. **Hyperparameter space not exhaustively searched:** One configuration per method; grid search could improve AUC by 0.02-0.05 pp.
3. **Advanced methods not compared:** CDAN (Long et al., 2018), MCD (Saito et al., 2018), Wasserstein DA - these may overcome GRL degeneracy in the heterogeneous setting.
4. **Temporal and geographic scope:** TDCCC (Taiwan, 2005) - application to modern systems (Vietnam 2020+) requires validation for concept drift and distribution shift.

6.6 Future Research Directions

1. **HDA with explicit label-shift handling:** CDAN (Long et al., 2018) with multilinear conditioning, or IWDAN (Tachet et al., 2020) with importance weighting to address $P_S(Y) \neq P_T(Y)$.
2. **Multi-source HDA:** Combine LC, Prosper, Freddie Mac, FICO for a richer, more representative source.
3. **Validation on Vietnamese credit data:** Domain gap with US/Taiwan sources may be even larger; feature availability lower.
4. **Complementary techniques:** Pseudo-labeling, source-target MixUp, gradient surgery (GradVac, PCGrad) to reduce gradient conflict between label loss and domain loss.

Conclusion

Transfer learning is not always beneficial. Explicit 3-case negative-transfer ablation must be a **mandatory** component of any transfer learning study in risk-critical domains such as credit scoring, where performance degradation translates directly into real financial losses. (Wang et al., 2019; Rosenstein et al., 2005)

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